

Advanced Calculus HW 8

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Question 1

(a)

$$\begin{aligned}f(x, y) &= e^{-x^2-y^2+4x} \\ \nabla f(x, y) &= ((-2+4)e^{-x^2-y^2+4x}, (-2y)e^{-x^2-y^2+4x}) \\ \nabla f(x, y) &= \vec{0} \iff (x, y) = (2, 0)\end{aligned}$$

Hence $(2, 0)$ is a critical point

Then using the second partials test, we need to compute

$$\begin{aligned}D &= f_{xx}(2, 0)f_{yy}(2, 0) - (f_{xy}(2, 0))^2 \\ f_{xx}(x, y) &= -2e^{-x^2-y^2+4x} + (-2x+4)^2e^{-x^2-y^2+4x} \\ f_{xx}(2, 0) &= -2e^4 \\ f_{yy}(x, y) &= -2e^{-x^2-y^2+4x} + 4y^2e^{-x^2-y^2+4x} \\ f_{yy}(2, 0) &= -2e^4 \\ f_{xy}(x, y) &= (-2x+4)(-2y)e^{-x^2-y^2+4x} \\ f_{xy}(2, 0) &= 0\end{aligned}$$

Hence $D = 4e^4 > 0$

Since $f_{xx}(2, 0) > 0$ as well, we have that there is a relative maximum of f at $(2, 0)$

(b)

$$f(x, y) = xy + \frac{2}{x} + \frac{4}{y}$$

Note: f is not defined for all points with $x = 0$ or $y = 0$

$$\begin{aligned}\nabla f(x, y) &= (y - \frac{2}{x^2}, x - \frac{4}{y^2}) \\ \nabla f(x, y) &= \vec{0} \iff x^2y = 2 \text{ and } xy^2 = 4\end{aligned}$$

The only such point is $(1, 2)$

Hence $(1, 2)$ is a critical point

Then using the second partials test, we need to compute

$$\begin{aligned}D &= f_{xx}(1, 2)f_{yy}(1, 2) - (f_{xy}(1, 2))^2 \\ f_{xx}(x, y) &= \frac{4}{x^3} \\ f_{xx}(1, 2) &= 4\end{aligned}$$

$$f_{yy}(x, y) = \frac{8}{y^3}$$

$$f_{yy}(1, 2) = 1$$

$$f_{xy}(x, y) = 1$$

Hence $D = 3 > 0$

Since $f_{xx}(1, 2) > 0$ as well, we have that there is a relative maximum of f at $(1, 2)$

(c)

$$f(x, y) = e^x \cos(y)$$

$$\nabla f(x, y) = (e^x \cos(y), -e^x \sin(y))$$

Since $e^x \neq 0 \forall x$ and $\cos(y) = \sin(y) \neq 0 \forall y$,

$$\nabla f(x, y) \neq \vec{0} \forall x, y$$

Hence, no critical points and hence no relative maxima, minima or saddle points

Question 2

Step 1

Find all interior critical points

$$\nabla f(x, y) = (y^2, 2yx)$$

$\nabla f(x, y) = \vec{0} \iff y = 0$ or $x = 0$, all such points are on the boundary, hence no interior critical points

Step 2

Find all boundary points on which relative extrema can occur

- $x \in [0, 1], y = 0$

$$f(x, 0) = 0$$

All such points are candidates

- $y \in [0, 1], x = 0$

$$f(0, y) = 0$$

All such points are candidates

- All points s.t. $x > 0, y > 0, x^2 + y^2 = 1$

$$x \in (0, 1), y = \sqrt{1 - x^2}$$

$$f(x, \sqrt{1 - x^2}) = x(1 - x^2) = x - x^3$$

$$\text{Extremised at } f'(x, \sqrt{1 - x^2}) = 0 \iff 1 - 3x^2 = 0$$

$$\text{Only at } x = \frac{1}{\sqrt{3}}$$

$$\text{Then at the point } \left(\frac{1}{\sqrt{3}}, \sqrt{\frac{2}{3}} \right)$$

Step 3

Evaluate all found points

$$f(x, 0) = 0 \forall x$$

$$f(0, y) = 0 \forall y$$

$$f\left(\frac{1}{\sqrt{3}}, \sqrt{\frac{2}{3}}\right) = \frac{2}{3\sqrt{3}}$$

Hence the absolute minima of this function on $\mathcal{R}$ is 0 and the absolute maximum is $\frac{2}{3\sqrt{3}}$

Question 3

As $P(1, -1, 1)$ lies on the plane $4x + 3y + z = 2$, the closest point on the plane to P is simply P

Using the method of Lagrange multipliers, we have the function to minimize as the square of a point to the point $P(1, -1, 1)$, $f(x) = (x - 1)^2 + (y + 1)^2 + (z - 1)^2$, and with $g(x) = 4x + 3y + z - 2$, the equation of the plane

Then we calculate

$$\nabla f(x, y, z) = (2(x - 1), 2(y + 1), 2(z - 1))$$

$$\nabla g(x, y, z) = (4, 3, 1)$$

Then, we find λ such that $\nabla f = \lambda \nabla g$

$$(2(x - 1), 2(y + 1), 2(z - 1)) = (4\lambda, 3\lambda, \lambda)$$

Solving these three linear equations we get $(x, y, z) = (1, -1, 1)$, as expected